

Predictability-Aware Sparse Subcarrier Hopping With Bayesian Reliability-Weighted Decoding Against Delayed Reactive Jamming

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Abstract—Sparse subcarrier hopping can evade a delayed reactive jammer, but hard recent-set exclusion can also make the next eligible set more predictable. This paper studies that tradeoff on a common OFDM grid through memory-constrained sparse hopping (MCSH), predictability-aware soft hopping (PASH), and Bayesian reliability weighting (BRW). MCSH selects the least recently used active bins and, if $(L_h + 1)S \leq N$, eliminates steady-state collisions from any follower whose delay does not exceed L_h . Stress tests reveal the cost of this hard rule: collision spikes can appear just outside the design memory and a candidate-aware jammer can exploit the reduced eligible pool. PASH therefore uses softened recent-use penalties and an optional inclusion-probability cap, trading perfect one-slot avoidance for lower worst-lag exposure. BRW marginalizes the unknown QPSK symbol under clean and jammed variance hypotheses, while an exact mixture-LLR decoder is used as a receiver benchmark. Across 251 baseline points and 60 adversarial stress points, MCSH reduces one-slot follower collision from approximately 0.125 to the acquisition-only level of 0.016. Under candidate-aware and best-lag attacks, however, MCSH+mixture-LLR has BLER 0.142 and 0.208, whereas capped PASH reduces them to 0.083 and 0.050, respectively. The results identify both a usable low-complexity design and the conditions under which it should not be claimed secure.

Index Terms—Anti-jamming communications, frequency hopping, OFDM, reactive jamming, reliability-weighted decoding, sparse multicarrier signaling.

I. INTRODUCTION

Reactive and spectrum-aware jammers challenge conventional frequency hopping (FH) because they sense the legitimate waveform and redirect interference toward its current or recent frequencies. Recent work addresses the problem using index-modulated FH [?], [1], data-driven FH-OFDM [2], reinforcement learning (RL) [3]–[5], and anti-jamming games [6]. Parallel work improves coded OFDM-IM [7], [8], reactive-jammer detection [9]–[11], and robust channel estimation [12].

A separate sequence-design literature studies wide-gap and no-hit-zone FH sequences [13], [14]. Those constructions establish valuable correlation and spacing guarantees, but a sparse multicarrier symbol activates a *set* of frequencies simultaneously. Its collision behavior, conditional pattern entropy, and interaction with coded soft decoding are not captured by a scalar FH sequence alone. Moreover, anti-jamming comparisons can conflate different physical bandwidths, power, coding, or jammer knowledge.

This paper starts from a controlled common-grid baseline and develops a joint transmitter–receiver method rather than claiming a universal FH processing gain. Its contributions are:

- A randomized, set-valued MCSH construction is formulated for sparse OFDM. A simple feasibility condition and

follower-collision law show exactly when an L_h -memory pattern removes delayed reactive collisions.

- PASH is introduced as a soft, capped alternative to hard exclusion. It directly targets the predictability failure found in lag-spectrum and candidate-aware jammer tests.
- BRW replaces binary erasure with a posterior effective-variance metric. It is compared with hard erasure, unweighted decoding, oracle erasure, and an exact clean/jammed mixture-LLR benchmark.
- The methods are ablated under blind partial-band, sweep, fixed-delay, random-delay, best-lag, and candidate-aware jammers. Additional experiments vary actual delay, spectral load, JSR, Pedestrian A/Vehicular A fading, CSI error, jammer-variance mismatch, and hopping-pattern synchronization error.
- Failure conditions are reported explicitly: hard MCSH can lose its advantage outside the design memory, capped PASH sacrifices known one-slot optimality, and no hopping rule defeats a zero-delay full-information jammer.

The contribution is not a claim of inventing no-hit-zone FH, nor a cryptographic security claim. It is a set-valued sparse-OFDM construction, collision/predictability analysis, and posterior-weighted coded receiver evaluated under a single reproducible physical-grid model.

II. RELATED WORK AND RESEARCH GAP

Table I summarizes the closest research directions. RL and game-based methods optimize decisions in dynamic environments, but require training, state design, or online interaction. Enhanced index-modulated FH specifically targets a power-correlated reactive jammer [1]. No-hit-zone sequence work provides strong scalar sequence constructions [14], whereas coded OFDM-IM and FH-OFDM erasure receivers demonstrate the importance of decoder cooperation [7], [8], [15]. The gap addressed here is a lightweight set-valued construction with an explicit delayed-follower guarantee and a non-binary reliability interface to the channel decoder.

III. COMMON-GRID SYSTEM AND JAMMER MODEL

A. Sparse OFDM Signal Model

Consider an allocated OFDM grid of N orthogonal physical bins. At symbol interval m , the transmitter activates a set $\mathcal{A}_m \subseteq \{0, \dots, N - 1\}$ with $|\mathcal{A}_m| = S$. The same number of unit-energy QPSK symbols, average power, symbol duration, system-band allocation, and rate-1/2 terminated convolutional code are used by all compared hopping patterns.

For active bin $k \in \mathcal{A}_m$, the ideal-FFT observation is

$$y_{k,m} = H_k x_{k,m} + \mathbb{1}\{k \in \mathcal{J}_m\} z_{k,m} + n_{k,m}, \quad (1)$$

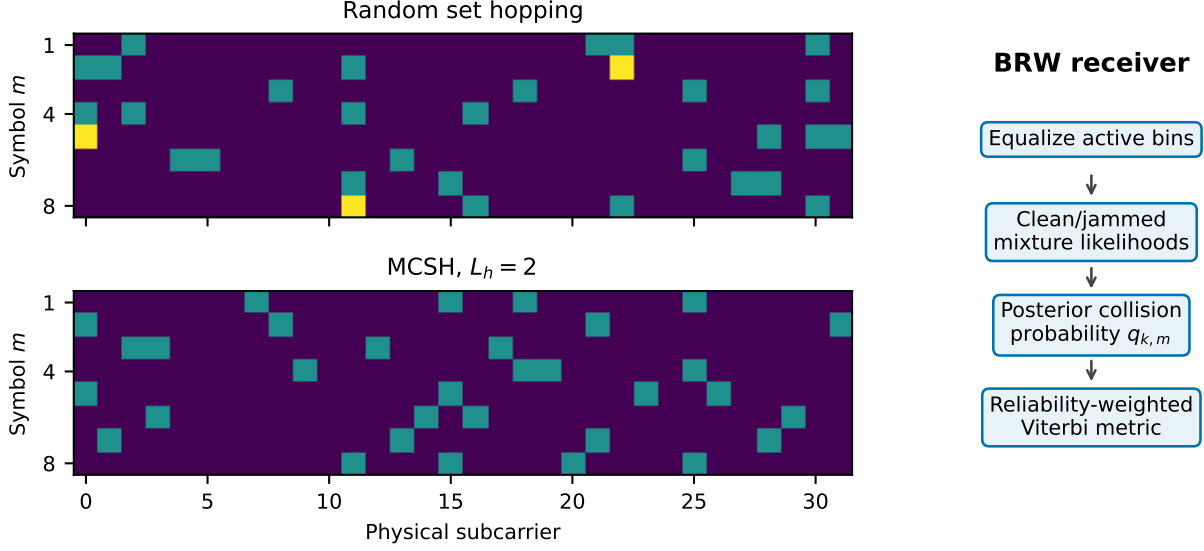


Fig. 1. MCSH suppresses recent-set reuse while retaining randomized choices. BRW maps clean/jammed mixture likelihoods to continuous Viterbi reliability weights. Repeated adjacent bins are highlighted in the random example.

TABLE I
POSITIONING RELATIVE TO REPRESENTATIVE APPROACHES

Approach	Main object	Reactive-jammer treatment	Coded soft reliability	Main distinction/limit
RL/game FH [4]–[6]	Policy/action	Learned or strategic response	Usually external	Adapts broadly; requires training/state interaction
Enhanced IM-FH [1]	Index modulation	Power-correlation resistance	Not the main focus	Targets a different reactive-jammer mechanism
Wide-gap/NHZ FHS [13], [14]	Scalar sequence	Correlation and no-hit-zone bounds	No	Does not model simultaneous sparse active-bin sets
Coded OFDM-IM/FH [7], [8], [15]	Code/erasure	Detection and decoding	Hard/oracle erasure common	No finite-memory set-collision guarantee
This work	Sparse active-bin set	Finite-memory exclusion	Posterior-weighted Viterbi	Limited by delay mismatch and pattern synchronization

where H_k is the communication-channel response, $n_{k,m} \sim \mathcal{CN}(0, N_0)$, and $\mathcal{J}_m$ is the jammed-bin set. The jammer has fixed total received energy $S 10^{\text{JSR}/10}$ per symbol, divided uniformly across $\mathcal{J}_m$. Define

$$c_m = \frac{|\mathcal{A}_m \cap \mathcal{J}_m|}{S}. \quad (2)$$

B. Jammer Classes

Four classes are evaluated. Blind partial-band jamming (PBJ) independently occupies a fraction ρ of the grid. A sweep jammer translates a contiguous block across symbols. A d -slot follower uses

$$\mathcal{J}_m = \mathcal{A}_{m-d}, \quad (3)$$

after a random acquisition phase of d symbols. A union follower attacks $\bigcup_{\ell=1}^{L_j} \mathcal{A}_{m-\ell}$. A zero-delay limit, $\mathcal{J}_m = \mathcal{A}_m$, is included as an intentionally unfavorable boundary rather than a scenario the proposed method claims to solve.

Independent random set hopping gives

$$\mathbb{E}[c_m] = \frac{S}{N} \quad (4)$$

against a single delayed follower. For blind PBJ, $\mathbb{E}[c_m] = \rho$ for both independent and memory-constrained patterns; therefore

MCSH should not be expected to create a blind-jammer mean-energy gain.

IV. SPARSE HOPPING WITH PREDICTABILITY CONTROL

A. Memory-Constrained Construction

For candidate physical bin k , define its recent-use count

$$r_{k,m} = \sum_{\ell=1}^{\min(L_h, m)} \mathbb{1}\{k \in \mathcal{A}_{m-\ell}\}. \quad (5)$$

MCSH selects the S bins having the smallest $r_{k,m}$, with pseudorandom tie-breaking generated from a shared seed. Thus, it first uses bins outside the recent union. If too few such bins exist, it reuses the least recently occupied bins. The implementation requires no online RL training or jammer feedback.

B. Collision Guarantee

Proposition 1: If $(L_h + 1)S \leq N$, MCSH produces

$$\mathcal{A}_m \cap \left(\bigcup_{\ell=1}^{L_h} \mathcal{A}_{m-\ell} \right) = \emptyset \quad (6)$$

for all valid m .

Proof: Assume the previous at most L_h sets satisfy the construction. Their union contains at most $L_h S$ bins, leaving at least $N - L_h S \geq S$ candidates with $r_{k,m} = 0$. MCSH selects S of these zero-count bins, proving (6). The initial step is immediate, so the property follows by induction. $\square$

Corollary 1: For a d -slot follower with $d \leq L_h$, post-acquisition collision is zero. If the first d jammer sets are independently random, the frame-average expectation for an M -symbol frame is

$$\mathbb{E}[\bar{c}_{\text{MCSH}}] = \frac{d}{M} \frac{S}{N}, \quad d \leq L_h, \quad (7)$$

versus S/N for independent random hopping. The same zero steady-state result holds for a union follower with $L_j \leq L_h$.

The guarantee has two important limits. First, it is infeasible when $(L_h + 1)S > N$. Second, an actual delay beyond L_h is unconstrained and can occasionally collide more than independent hopping. The delay and load experiments in Section VII test both limits.

C. Predictability-Aware Soft Hopping

Hard exclusion is optimal against a known $d \leq L_h$ follower, but it exposes a smaller next-symbol candidate pool. To reduce this predictability, PASH draws the active set without replacement using softened recent-use weights

$$w_{k,m} = p_{\text{floor}} + \exp\left(-\alpha \sum_{\ell=1}^H \mathbb{1}\{k \in \mathcal{A}_{m-\ell}\}\right), \quad (8)$$

where H is the soft memory, α controls recent-use repulsion, and $p_{\text{floor}} > 0$ prevents deterministic exclusion. The capped variant further limits the approximate conditional inclusion probability,

$$p_{k,m} = \min\{p_{\text{max}}, \eta_m w_{k,m}\}, \quad \sum_{k=1}^N p_{k,m} = S, \quad (9)$$

with η_m found by bisection. The final set is sampled according to these weights. Thus, PASH-cap deliberately gives up zero short-delay collision in exchange for lower worst-lag and candidate-aware exposure.

D. Randomness and Complexity

When the recent sets are disjoint, the conditional number of next-set choices is $\binom{N-L_h S}{S}$, giving

$$\mathcal{H}_m = \log_2 \binom{N-L_h S}{S}. \quad (10)$$

For $N = 512$, $S = 64$, and $L_h = 2$, this is approximately 245 bits per symbol. This is an unpredictability measure under pseudorandom tie-breaking, not a cryptographic-security proof. Recent-use counting costs $O(L_h S + N)$; selecting the lowest-count bins can be implemented in $O(N \log S)$. PASH adds one weight pass, a scalar cap search if (9) is used, and weighted sampling without replacement; no training phase or jammer feedback is required.

V. BAYESIAN RELIABILITY-WEIGHTED DECODING

A. Posterior Collision Probability

After equalization, let $\tilde{y}_{k,m} = y_{k,m}/\hat{H}_k$. BRW uses

$$\mathcal{H}_0 : \tilde{y} = x + \tilde{n}, \quad \tilde{n} \sim \mathcal{CN}(0, v_0), \quad (11)$$

$$\mathcal{H}_1 : \tilde{y} = x + \tilde{n} + \tilde{z}, \quad \tilde{n} + \tilde{z} \sim \mathcal{CN}(0, v_1), \quad (12)$$

where $v_1 > v_0$. Unlike a nearest-symbol residual test, the unknown QPSK symbol is marginalized:

$$p_i(\tilde{y}) = \frac{1}{4\pi v_i} \sum_{x \in \mathcal{X}_{\text{QPSK}}} \exp\left(-\frac{|\tilde{y} - x|^2}{v_i}\right). \quad (13)$$

With prior collision probability π , the posterior is

$$q = \Pr(\mathcal{H}_1 | \tilde{y}) = \frac{\pi p_1(\tilde{y})}{\pi p_1(\tilde{y}) + (1 - \pi) p_0(\tilde{y})}. \quad (14)$$

Noise variance can be obtained from conventional estimators, while jammer variance is estimated from sensed interference power. Section VII explicitly varies jammer-variance mismatch.

B. Weighted Viterbi Metric

BRW forms the effective variance

$$v_{\text{eff}} = (1 - q)v_0 + qv_1 \quad (15)$$

and weights each coded-bit branch metric by $1/v_{\text{eff}}$. For trellis branch b with expected antipodal coded components μ_b ,

$$\Lambda_b = \sum_j \frac{(s_j - \mu_{b,j})^2}{v_{\text{eff},j}}, \quad (16)$$

where s_j is the QPSK soft component. Hard erasure is recovered by setting weights to zero when $q > 0.5$; oracle erasure uses the true collision mask only as an upper bound. In the adversarial stress suite we also report an exact clean/jammed mixture-LLR Viterbi metric that evaluates the two QPSK-mixture likelihoods for each bit. It is used to avoid overstating PASH or MCSH effects that are actually caused by receiver suboptimality.

BRW adds eight QPSK distance/exponential terms per active sample and $O(SM)$ storage for posterior weights. The constraint-length-7 Viterbi decoder remains dominant with complexity proportional to $2^{K-1} SM$.

VI. EXPERIMENTAL METHODOLOGY

All experiments are generated by an independent Python implementation and do not reuse the invalid legacy MATLAB results identified during the project audit. Table II lists the default configuration.

The core study separates pattern and receiver effects using random+K7, random+BRW, MCSH+K7, MCSH+hard erasure, MCSH+BRW, and MCSH+oracle erasure. The 251-point suite contains 63,000 frames and approximately 32.8 million information bits. It additionally varies follower delay $d = 1, \dots, 6$, memory $L_h \in \{1, 2, 4, 6\}$, JSR from -5 to 15 dB, active load, channel profile, CSI NMSE, jammer-variance error, and erroneous-pattern probability. A pattern desynchronization event shifts all receiver bin indices for that OFDM symbol,

TABLE II
DEFAULT EXPERIMENTAL CONFIGURATION

Parameter	Value
Physical grid / active bins	$N = 512, S = 64$
Symbols per frame	$M = 8$
Modulation and code	QPSK; terminated rate-1/2 K7 (171, 133) ₈
Default SNR / JSR	$E_s/N_0 = 12$ dB / JSR=5 dB
PBJ / sweep occupancy	$0.25N / 0.125N$
Sweep step	24 bins per symbol
MCSH memory	$L_h = 2$ unless varied
Channels	AWGN, Pedestrian A, Vehicular A
Core / robustness frames	300 / 200 per point
Adversarial stress	30 seeds $\times$ 5000 symbols; 240 frames per coded point
Uncertainty reporting	Wilson 95% BER confidence intervals

which models a severe mapping error rather than a mild phase offset.

The adversarial suite adds two tests that are closer to a publishable stress-test standard. First, continuous long-run hopping sequences over 30 seeds and 5000 OFDM symbols per seed estimate the lag-collision spectrum, worst-lag collision, candidate-aware collision, and bin-usage coefficient of variation. Second, 60 coded points report BLER and collision rate for random, MCSH, PASH, and capped PASH under fixed one-slot, fixed three-slot, random delay, best-lag, and candidate-aware jammers. Each point compares no weighting, BRW, and exact mixture-LLR decoding.

Finally, model-validity checks verify structural invariants before interpreting BER or BLER: random hopping matches the closed-form load S/N , MCSH has zero lag-1 and lag-2 collision for $L_h = 2$, worst-lag spikes occur at $L_h + 1$, and PASH-cap reduces the maximum of worst-lag and candidate-aware exposure while sacrificing zero short-delay collision. All nine checks pass.

VII. RESULTS AND DISCUSSION

A. Separating Pattern and Receiver Gains

Figure 2(a) shows that MCSH alone does not improve sweep-jammer mean collision or BER, as expected because the sweep is unrelated to recent active sets. BRW, however, reduces the sweep BER from approximately 10^{-2} to the 10^{-4} range at 12 dB. This is a receiver gain, not an MCSH gain. Blind PBJ gives the same conclusion: the pattern does not lower mean collision, whereas BRW retains useful soft information that hard erasure can discard.

For one- and two-slot followers, MCSH changes the collision mechanism. At $S/N = 1/8$, independent hopping collides at approximately 0.125. With $L_h = 2$, measured frame-average collision is approximately 0.016 for $d = 1$ and 0.032 for $d = 2$, agreeing with (7). The remaining collisions arise only from the deliberately random acquisition symbols. The union-of-two follower is similarly reduced from roughly 0.21 to 0.015.

B. Memory Depth and Feasibility Boundary

Figure 3(a) verifies the acquisition-only law for every $d \leq L_h$. It also shows the cost of mismatch. For example, $L_h = 2$ has higher collision than random hopping at $d = 3$ because the construction only controls the two most recent sets.

Increasing memory protects a larger delay range, but reduces the conditional choice count in (10).

The load sweep in Fig. 3(c) tests $(L_h + 1)S \leq N$. MCSH remains near its acquisition floor through $S = 160$ for $N = 512$, but collision rises sharply at $S = 192$, beyond the $S/N = 1/3$ feasibility boundary. This prevents the zero-collision property from being overgeneralized to heavily loaded multicarrier systems.

C. Adversarial Predictability Stress Tests

Figure 4 addresses the main weakness of hard recent-set exclusion. Long-run structural tests show that MCSH with $L_h = 2$ has zero mean collision for lags 1–2, but its worst-lag collision rises to 0.167 at lag 3. Increasing to $L_h = 6$ shifts the problem rather than removing it: the worst-lag collision reaches 0.500 at lag 7 because the eligible pool is extremely concentrated. These results explain why a delay mismatch can reverse the MCSH gain.

PASH-cap changes the design objective from “zero collision at selected short lags” to “bounded predictability over many plausible lags.” Its mean collision over lags 1–2 is 0.099, so it is worse than MCSH against a perfectly known one-slot follower. However, its worst-lag collision is 0.130 and its candidate-aware collision is 0.148, both lower than the MCSH $L_h = 2$ values of 0.167 and 0.167. In coded stress tests with mixture-LLR decoding, this tradeoff is visible: MCSH has BLER 0.004 for a fixed one-slot follower, but 0.125, 0.208, and 0.142 for fixed three-slot, best-lag, and candidate-aware jammers. PASH-cap gives 0.029 for the one-slot case, yet reduces those three stronger attacks to 0.013, 0.050, and 0.083. The result is not that PASH-cap dominates MCSH; it is that the correct hopping rule depends on the uncertainty and adversarial knowledge of the jammer delay.

D. JSR, Fading, CSI, and Synchronization

Figure 5(a) shows different roles for the two modules. MCSH reduces the number of follower collisions, while BRW prevents the remaining high-JSR samples from dominating the decoder. In Pedestrian A and Vehicular A fading, Fig. 5(b) shows that BRW continues to improve both random and memory-constrained hopping under perfect CSI. With Vehicular A channel-estimation error, performance degrades gradually through -10 dB NMSE rather than collapsing immediately.

Hopping synchronization is more critical. At a 1% erroneous-pattern rate, the MCSH+BRW BER rises into the 10^{-3} range, and at 5% it reaches the 2.576×10^{-2} level. This experiment uses a full-bin mapping error, so it should be read as a synchronization-failure bound. A deployable system requires an authenticated pattern resynchronization mechanism.

E. Receiver Ablation and Failure Boundaries

Figure 6(a) compares no weighting, hard erasure, BRW, and oracle erasure under sweep jamming. BRW consistently improves upon hard erasure because uncertain samples are

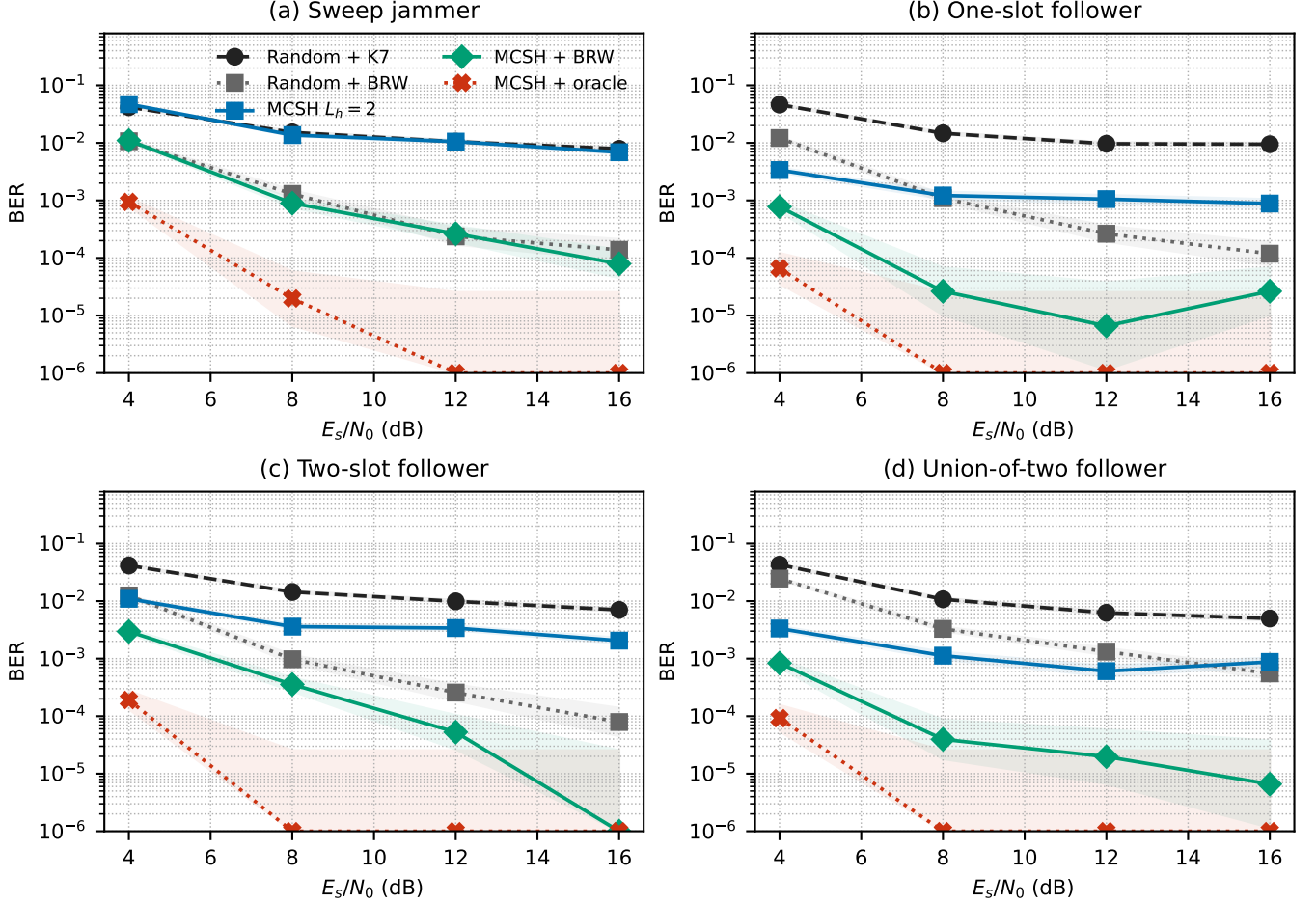


Fig. 2. Core BER results at JSR=5 dB. Curves at the 10^{-6} plotting floor with zero measured errors must be interpreted through their Wilson upper confidence bounds, not as zero true BER.

TABLE III
REPRESENTATIVE RESULTS AT $E_s/N_0 = 12$ dB AND JSR=5 dB

Jammer / condition	Scheme	Collision rate	BER	Wilson 95% interval
Sweep	MCSH+K7	0.125	1.053×10^{-2}	$[1.003, 1.106] \times 10^{-2}$
Sweep	MCSH+hard erasure	0.125	4.216×10^{-4}	$[3.302, 5.383] \times 10^{-4}$
Sweep	MCSH+BRW	0.125	2.635×10^{-4}	$[1.935, 3.588] \times 10^{-4}$
One-slot follower	random+K7	0.124	9.677×10^{-3}	$[9.197, 10.182] \times 10^{-3}$
One-slot follower	MCSH+K7	0.016	1.054×10^{-3}	$[0.903, 1.230] \times 10^{-3}$
One-slot follower	MCSH+BRW	0.016	6.588×10^{-6}	$[1.163, 37.317] \times 10^{-6}$
Union-of-two follower	random+K7	0.206	6.258×10^{-3}	$[5.874, 6.668] \times 10^{-3}$
Union-of-two follower	MCSH+BRW	0.015	1.976×10^{-5}	$[0.672, 5.811] \times 10^{-5}$

downweighted instead of discarded. The remaining oracle gap identifies posterior modeling and variance estimation as future receiver opportunities. Figure 6(b) shows that BRW remains useful under several decibels of jammer-power estimation error, though the best point need not occur at zero mismatch because the approximate two-component model does not represent every sweep sample exactly.

Finally, Fig. 6(c) gives the necessary negative result: when the jammer instantaneously occupies $\mathcal{A}_m$, all compared schemes approach BER 0.5; MCSH+BRW measures 0.494. MCSH is a defense against delayed followers, not a solution to a zero-latency full-information jammer.

F. Complexity and Practical Interpretation

MCSH adds a recent-use array and a partial selection per OFDM symbol. BRW adds constant-order mixture likelihoods per active sample, while the existing Viterbi decoder remains asymptotically dominant. No-jam experiments show no systematic BER penalty from the MCSH pattern itself. The main implementation cost is therefore pattern synchronization and variance estimation rather than decoder order.

Compared with RL-based approaches, MCSH/PASH-BRW has no training phase and the MCSH delayed-follower guarantee is explicit. Conversely, RL or game-based policies can adapt to richer jammer behavior that violates the assumed

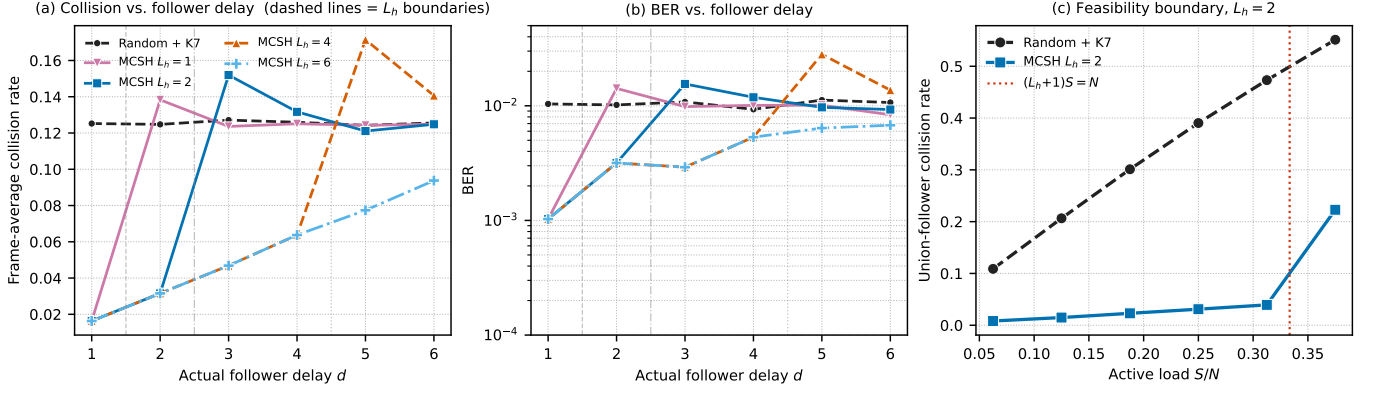


Fig. 3. Memory and load ablation at $E_s/N_0 = 12$ dB and JSR=5 dB. A memory chosen shorter than the actual delay can be worse than random hopping. The vertical line marks the zero-recent-overlap feasibility boundary for $L_h = 2$.

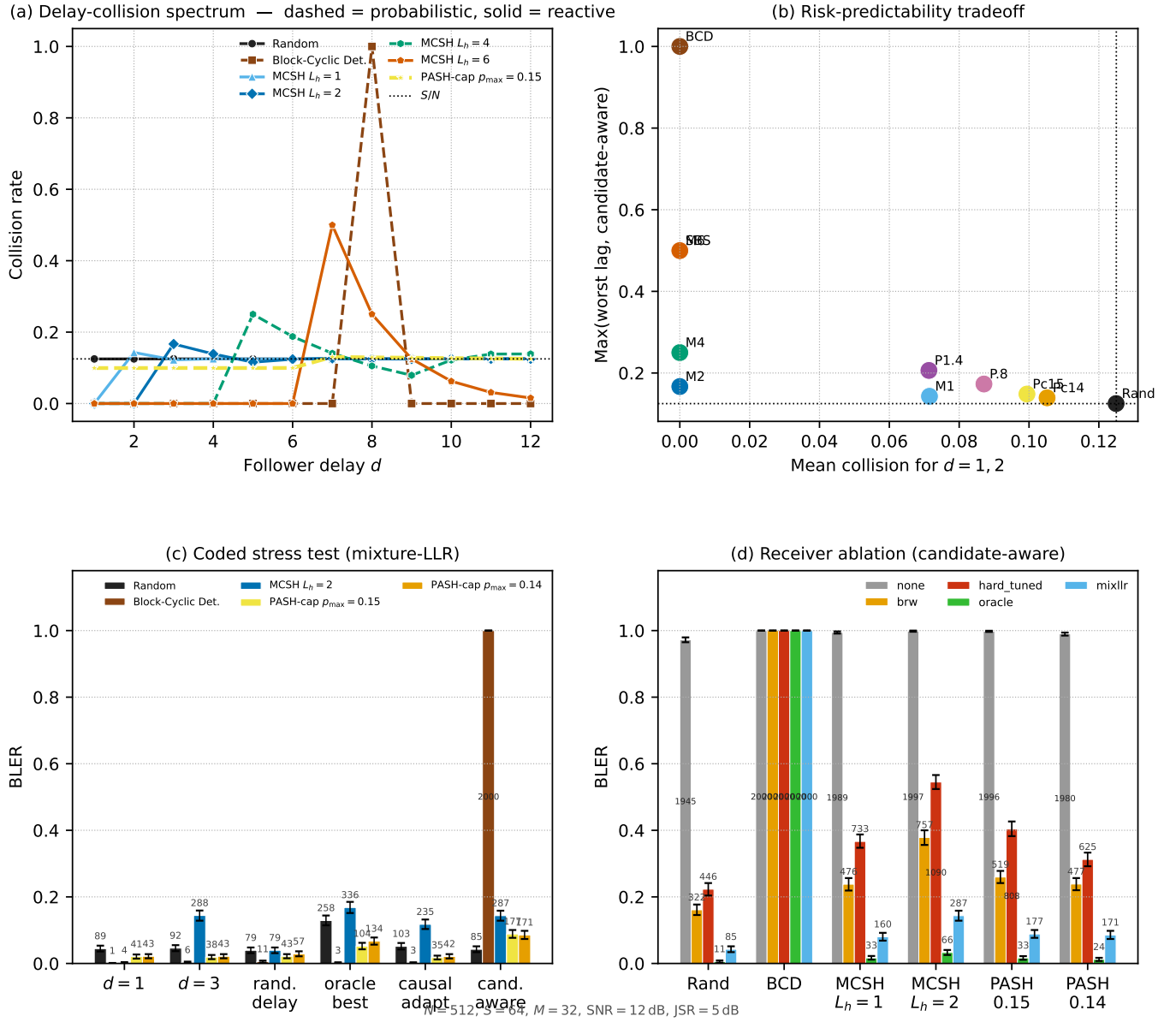


Fig. 4. Adversarial predictability stress tests. MCSH removes collisions inside its memory but creates lag spikes just outside it; capped PASH lowers worst-lag and candidate-aware exposure, then converts the structural gain into lower BLER under best-lag and candidate-aware jammers.

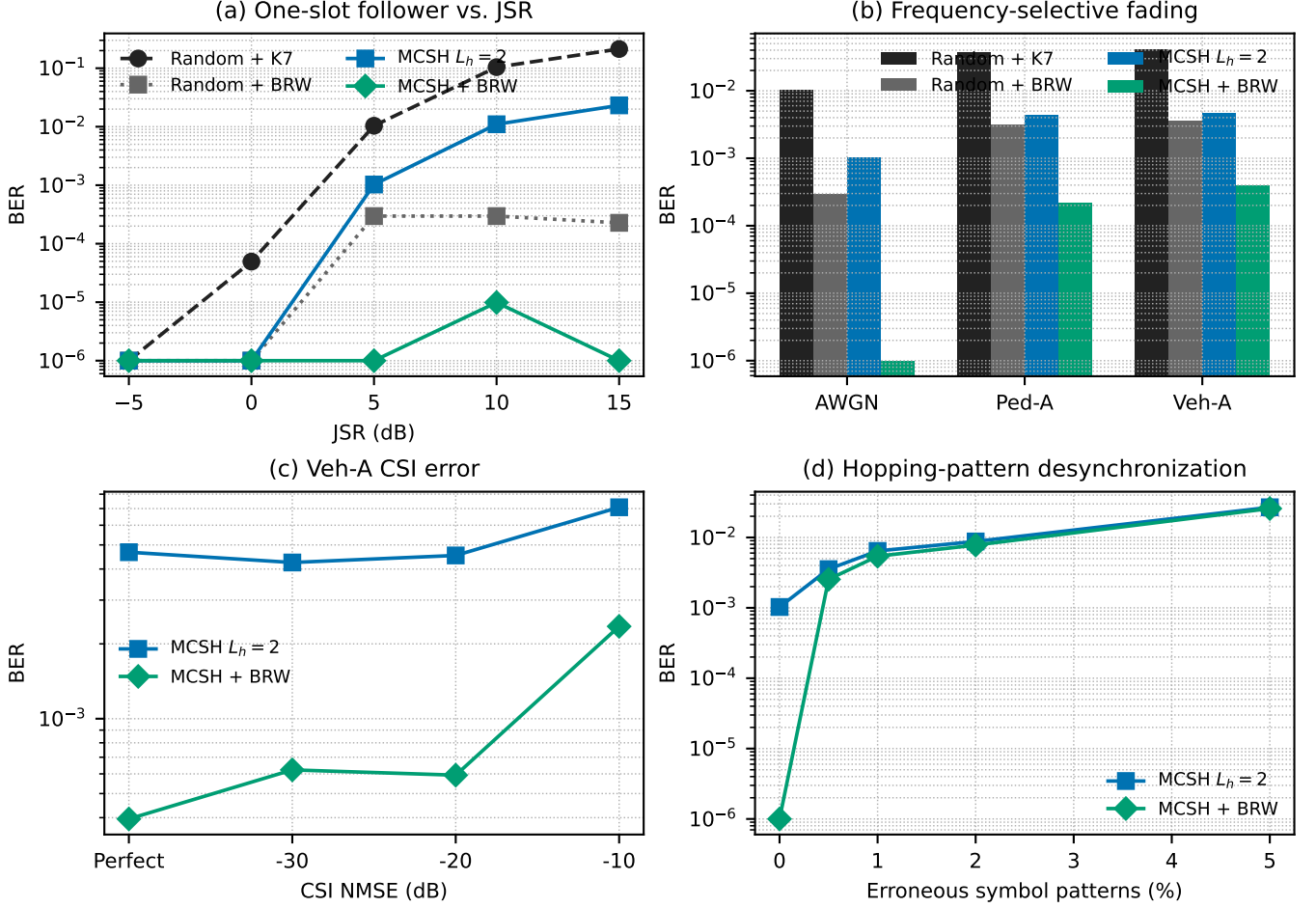


Fig. 5. Robustness experiments for a one-slot follower unless otherwise noted: JSR, multipath fading, CSI error, and hopping-pattern desynchronization.

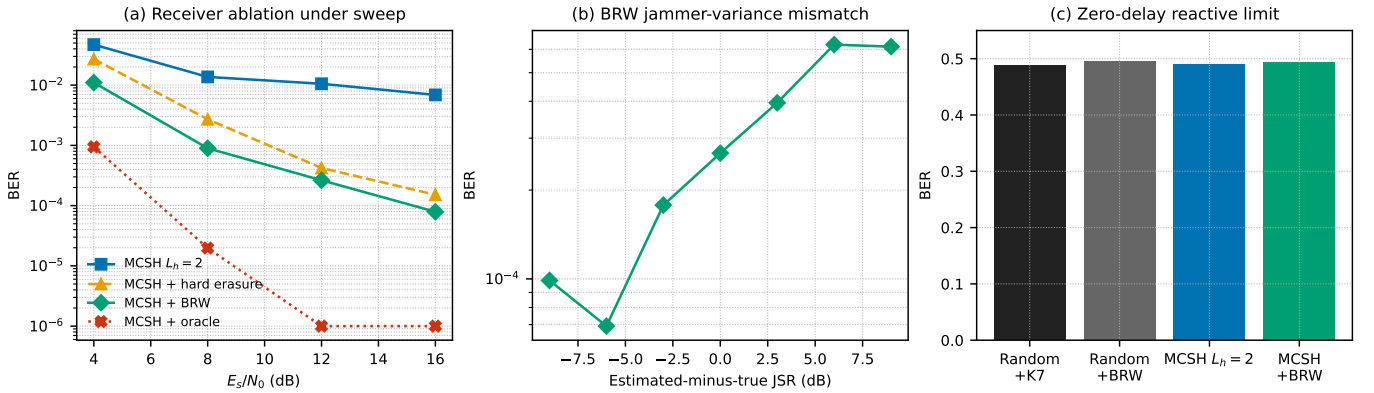


Fig. 6. Receiver ablation, jammer-variance mismatch, and the zero-delay reactive-jammer limit.

delay model. The methods are therefore complementary rather than directly ranked by a single BER number.

VIII. LIMITATIONS AND FUTURE WORK

The study uses an ideal FFT boundary and frequency-domain jammer model. Although Pedestrian A/Vehicular A fading, CSI error, and mapping desynchronization are included, timing offset, CFO-induced intercarrier interference,

pilot overhead, channel coding beyond K7, and hardware nonlinear effects remain open. The stronger best-lag and candidate-aware tests are still heuristic adversaries, not a full game-theoretic optimum. Likewise, PASH-cap is an interpretable capped exponential rule rather than a proven min-max hopping law. Future work should combine delay-belief estimation, adaptive memory, authenticated resynchronization, stronger codes, and over-the-air validation.

IX. CONCLUSION

Sparse subcarrier hopping against delayed reactive jamming is a joint collision, predictability, and soft-reliability problem. Under a feasible spectral load, MCSH removes steady-state collisions from followers within its design memory; under uncertain or adversarially selected delay, PASH-cap lowers worst-lag and candidate-aware exposure at the cost of known one-slot optimality. BRW and mixture-LLR decoding then show how much of that structural gain survives in a coded receiver. The negative and robustness tests identify the main boundaries: memory mismatch, excessive active load, zero jammer delay, and hopping desynchronization. These boundaries make the reported gains more reproducible and technically defensible.

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